

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

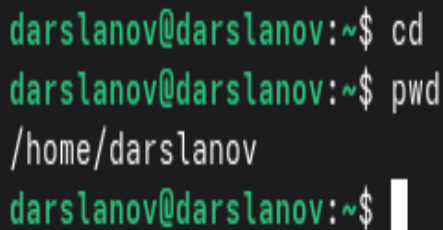
Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы



```
darslanov@darslanov:~$ cd  
darslanov@darslanov:~$ pwd  
/home/darslanov  
darslanov@darslanov:~$
```

A terminal window with a dark background and green text. It shows a user named 'darslanov' at a machine named 'darslanov' in the home directory '~'. The user enters 'cd' and then 'pwd', which outputs '/home/darslanov'. The prompt returns to '~\$'.

Рисунок 1: Путь к домашнему каталогу

Опции команды ls

```
darslanov@darslanov:~$ cd /tmp
darslanov@darslanov:/tmp$ ls
01edf5bb-cbf6-4009-9bd6-3fee66e7b82d.zip
2e1e298c-7d11-4b0b-ae5a-87a5bb4a78d5.zip
happd.sock
hsperfdata_darslanov
node-compile-cache
par-646172736c616e6f76
snap-private-tmp
systemd-private-a80113b2e236483ca1b57eaa4e457c22-abrtd.service-r0bLNn
systemd-private-a80113b2e236483ca1b57eaa4e457c22-chrond.service-gSx1vt
systemd-private-a80113b2e236483ca1b57eaa4e457c22-colord.service-mCCXx2
systemd-private-a80113b2e236483ca1b57eaa4e457c22-dbus-broker.service-cQsACs
systemd-private-a80113b2e236483ca1b57eaa4e457c22-fwupd.service-yRePj3
systemd-private-a80113b2e236483ca1b57eaa4e457c22-irqbalance.service-Mo4FWA
systemd-private-a80113b2e236483ca1b57eaa4e457c22-low-memory-monitor.service-8xi1MD
systemd-private-a80113b2e236483ca1b57eaa4e457c22-ModemManager.service-6gDaZ0
systemd-private-a80113b2e236483ca1b57eaa4e457c22-passim.service-F0Pe2L
systemd-private-a80113b2e236483ca1b57eaa4e457c22-pcsd.service-38B5Xc
systemd-private-a80113b2e236483ca1b57eaa4e457c22-polkit.service-W9tfBW
systemd-private-a80113b2e236483ca1b57eaa4e457c22-rtkit-daemon.service-1C4taJ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-switcheroo-control.service-TmYNz8
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-logind.service-DqBziZ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-resolved.service-qQ2KiQ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-upower.service-E0wEH7
VMwareDnD
vmware-root_895-3979642976
darslanov@darslanov:/tmp$
```

Рисунок 2: Команда ls

```
darсланов@darсланов:/tmp$ ls -la
.
..
01edf5bb-cbf6-4009-9bd6-3fee66e7b82d.zip
2e1e298c-7d11-4b0b-ae5a-87a5bb4a78d5.zip
.font-unix
happd.sock
hsperfdata_darсланов
.ICE-unix
node-compile-cache
par-646172736c616e6f76
snap-private-tmp
systemd-private-a80113b2e236483ca1b57eaa4e457c22-abrt.service-r0bLNn
systemd-private-a80113b2e236483ca1b57eaa4e457c22-chrond.service-gSx1vt
systemd-private-a80113b2e236483ca1b57eaa4e457c22-colord.service-mCCXx2
systemd-private-a80113b2e236483ca1b57eaa4e457c22-dbus-broker.service-cQsACs
systemd-private-a80113b2e236483ca1b57eaa4e457c22-fwupd.service-yRePj3
systemd-private-a80113b2e236483ca1b57eaa4e457c22-irqbalance.service-Mo4FWA
systemd-private-a80113b2e236483ca1b57eaa4e457c22-low-memory-monitor.service-8x11MD
systemd-private-a80113b2e236483ca1b57eaa4e457c22-ModemManager.service-6gDaZ0
systemd-private-a80113b2e236483ca1b57eaa4e457c22-passim.service-F0Pe2L
systemd-private-a80113b2e236483ca1b57eaa4e457c22-pcsd.service-38B5Xc
systemd-private-a80113b2e236483ca1b57eaa4e457c22-polkit.service-W9tfBW
systemd-private-a80113b2e236483ca1b57eaa4e457c22-rtkit-daemon.service-1C4taJ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-switcheroo-control.service-TmYNz8
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-logind.service-DqBZiZ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-resolved.service-qQ2KiQ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-upower.service-E0wEH7
VMwareDnD
vmware-root_895-3979642976
.X0-lock
```

Рисунок 3: Команда ls -la

```
darсланов@darсланов:/tmp$  
darсланов@darсланов:/tmp$ ls -l  
total 7744  
-rw-r--r--. 1 darсланов darсланов 3961995 Sep  4 11:12 01edf5bb-cbf6-4009-9bd6-3fee66e7b82d.zip  
-rw-r--r--. 1 darсланов darсланов 3961995 Sep  4 11:13 2e1e298c-7d11-4b0b-ae5a-87a5bb4a78d5.zip  
srw-rw-rw-. 1 root      root          0 Sep  4 11:11 happd.sock  
drwxr-xr-x. 2 darсланов darсланов    40 Sep  4 11:28 hsperfdata_darсланов  
drwxr-xr-x. 3 darсланов darсланов   60 Sep  4 11:34 node-compile-cache  
drwx-----. 3 darсланов darсланов   60 Sep  4 11:24 par-646172736c616e6f76  
drwx-----. 2 root      root        40 Sep  4 11:10 snap-private-tmp  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-abrt.service-r0bLnn  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-chronyd.service-gSxivt  
drwx-----. 3 root      root        60 Sep  4 11:11 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-colord.service-mCCXx2  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-dbus-broker.service-cQsACs  
drwx-----. 3 root      root        60 Sep  4 11:11 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-fwupd.service-yRePj3  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-irqbalance.service-Mo4FWA  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-low-memory-monitor.service-8xi1MD  
drwx-----. 3 root      root        60 Sep  4 11:10 systemd-private-a80113b2e236483ca1b57eaa4e457c22  
-ModemManager.service-6gDaZ0  
drwx-----. 3 root      root        60 Sep  4 11:11 systemd-private-a80113b2e236483ca1b57eaa4e457c22
```

Рисунок 4: Команда ls -l

```
darсланов@darсланов:/tmp$ ls -f
.
..
node-compile-cache
hsperfdata_darсланов
par-646172736c616e6f76
VMwareDnD
2e1e298c-7d11-4b0b-ae5a-87a5bb4a78d5.zip
01edf5bb-cbf6-4009-9bd6-3fee66e7b82d.zip
systemd-private-a80113b2e236483ca1b57eaa4e457c22-passim.service-F0Pe2L
systemd-private-a80113b2e236483ca1b57eaa4e457c22-fwupd.service-yRePj3
.X1-lock
.X0-lock
systemd-private-a80113b2e236483ca1b57eaa4e457c22-pcsd.service-38B5Xc
systemd-private-a80113b2e236483ca1b57eaa4e457c22-colord.service-mCCXx2
happd.sock
systemd-private-a80113b2e236483ca1b57eaa4e457c22-ModemManager.service-6gDaZ0
vmware-root_895-3979642976
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-logind.service-DqBziZ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-upower.service-E0wEH7
systemd-private-a80113b2e236483ca1b57eaa4e457c22-switcheroo-control.service-TmYNz8
systemd-private-a80113b2e236483ca1b57eaa4e457c22-rtkit-daemon.service-1C4taJ
systemd-private-a80113b2e236483ca1b57eaa4e457c22-polkit.service-W9tfBW
systemd-private-a80113b2e236483ca1b57eaa4e457c22-low-memory-monitor.service-8xi1MD
systemd-private-a80113b2e236483ca1b57eaa4e457c22-chrond.service-gSx1vt
systemd-private-a80113b2e236483ca1b57eaa4e457c22-irqbalance.service-Mo4FWA
systemd-private-a80113b2e236483ca1b57eaa4e457c22-abrt.service-r0bLNN
systemd-private-a80113b2e236483ca1b57eaa4e457c22-dbus-broker.service-cQsACs
systemd-private-a80113b2e236483ca1b57eaa4e457c22-systemd-resolved.service-qQ2K1Q
```

Рисунок 5: Команда ls -f

```
darslanov@darslanov:/tmp$ cd /var/spool/  
darslanov@darslanov:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron  
drwx-----. 1 root root   18 Oct 23 2025 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp      6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail  276 Sep  4 11:10 mail  
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth  
darslanov@darslanov:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
darсланov@darсланov:~/var/spool$ cd
darсланov@darсланov:~/var/spool$ ls
Desktop Documents Downloads git-extended Music Pictures Public Templates Videos work
darсланov@darсланov:~/var/spool$ ls -al
total 24
drwx----- 1 darсланov darсланov 370 Sep  4 11:46 .
drwxr-xr-x 1 root      root      276 Sep  4 11:10 ..
-rw----- 1 darсланov darсланov 1771 Sep  4 11:48 .bash_history
-rw-r--r-- 1 darсланov darсланov  18 Jul 23  2025 .bash_logout
-rw-r--r-- 1 darсланov darсланov 144 Jul 23  2025 .bash_profile
-rw-r--r-- 1 darсланov darсланov 679 Sep  4 11:34 .bashrc
drwx----- 1 darсланov darсланov 484 Sep  4 11:39 .cache
drwx----- 1 darсланov darсланov 328 Sep  4 11:39 .config
drwxr-xr-x 1 darсланov darсланov   0 Sep  4 11:11 Desktop
drwxr-xr-x 1 darсланov darсланov   0 Sep  4 11:11 Documents
drwxr-xr-x 1 darсланov darсланov   0 Sep  4 11:11 Downloads
-rw-r--r-- 1 darсланov darсланov 334 Nov 14  2025 .emacs
-rw-r--r-- 1 darсланov darсланov 234 Sep  4 11:46 .gitconfig
drwxr-xr-x 1 darсланov darсланov  74 Sep  4 11:42 git-extended
drwx----- 1 darсланov darсланov 136 Sep  4 11:18 .gnupg
drwx----- 1 darсланov darсланov  20 Sep  4 11:11 .local
drwxr-xr-x 1 darсланov darсланov  34 Oct 23  2025 .mozilla
```

Рисунок 7: Файлы в домашнем каталоге

```
darslanov@darslanov:~$  
darslanov@darslanov:~$ mkdir newdir  
darslanov@darslanov:~$ mkdir newdir/morefun  
darslanov@darslanov:~$ mkdir letters memos misk  
darslanov@darslanov:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
darslanov@darslanov:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
darslanov@darslanov:~$ rm -r letters/ memos/ misk/  
darslanov@darslanov:~$ rm -r newdir/  
darslanov@darslanov:~$ ls  
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work  
darslanov@darslanov:~$
```

Рисунок 8: Действия с каталогами

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':  
solway.jpg  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':  
:  
csl tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/  
csl':  
gost-r-7-0-5-2008-numeric.csl  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/  
tex':  
preamble.tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':  
image-report mpv-shot  
darslanov@darslanov:~$ ls -t  
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos  
darslanov@darslanov:~$
```

Рисунок 9: Команда ls -R и ls -t

```
darslanov@darslanov:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

Change the current directory to DIR.  The default DIR is the value of the
HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR.  Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory.  If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option `cdable_vars' is set,
the word is assumed to be a variable name.  If that variable has a value,
its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.
```

Рисунок 10: Справка по команде cd

```
PWD(1)                                User Commands                                PWD(1)

NAME
  pwd - print name of current/working directory

SYNOPSIS
  pwd [OPTION]...

DESCRIPTION
  Print the full filename of the current working directory.

  -L, --logical
        use PWD from environment, even if it contains symlinks

  -P, --physical
        resolve all symlinks

  --help display this help and exit

  --version
        output version information and exit

  If no option is specified, -P is assumed.

  Your shell may have its own version of pwd, which usually supersedes the version described
  here. Please refer to your shell's documentation for details about the options it supports.

AUTHOR
  Written by Jim Meyering.

REPORTING BUGS
  GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
  Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help display this help and exit

    --version
        output version information and exit

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
  rmdir - remove empty directories

SYNOPSIS
  rmdir [OPTION]... DIRECTORY...

DESCRIPTION
  Remove the DIRECTORY(ies), if they are empty.

  --ignore-fail-on-non-empty
    ignore each failure to remove a non-empty directory

  -p, --parents
    remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

  -v, --verbose
    output a diagnostic for every directory processed

  --help display this help and exit

  --version
    output version information and exit

AUTHOR
  Written by David MacKenzie.

REPORTING BUGS
  GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
  Report any translation bugs to <https://translationproject.org/team/>

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default,
  it does not remove directories.

  If the -I or --interactive=once option is given, and there are more than three files or the -r,
-R, or --recursive are given, then rm prompts the user for whether to proceed with the entire
  operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option
  is not given, or the -i or --interactive=always option is given, rm prompts the user for
  whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
      ignore nonexistent files and arguments, never prompt

  -i
      prompt before every removal

  -I
      prompt once before removing more than three files, or when removing recursively; less
      intrusive than -i, while still giving protection against most mistakes

  --interactive[=WHEN]

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
149 cd /var/spool/
150 ls -l
151 cd
152 ls
153 ls -al
154 mkdir newdir
155 mkdir newdir/morefun
156 mkdir letters memos misk
157 ls
158 rm letters/ memos/ misk/
159 rm -r letters/ memos/ misk/
160 rm -r newdir/
161 ls
162 ls -R
163 ls -t
164 help cd
165 man pwd
166 man mkdir
167 man rmdir
168 man rm
169 history
darslanov@darslanov:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.